

## 4. Linear regression

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## Model notation

**Likelihood:**  $W \sim \text{Binomial}(N, p)$

**Prior:**  $p \sim \text{Uniform}(0, 1)$

- Likelihoods put probability distributions on the data
- Priors put probability distributions on the parameters (!!!)

# Building a model of human height

```
1 library(rethinking)
2 data("Howell1")
3 d<-Howell1
4 # subset out children
5 d2 <- d |>
6   filter(age>=18)
7 head(d2)
```

|   | height  | weight   | age | male |
|---|---------|----------|-----|------|
| 1 | 151.765 | 47.82561 | 63  | 1    |
| 2 | 139.700 | 36.48581 | 63  | 0    |
| 3 | 136.525 | 31.86484 | 65  | 0    |
| 4 | 156.845 | 53.04191 | 41  | 1    |
| 5 | 145.415 | 41.27687 | 51  | 0    |
| 6 | 163.830 | 62.99259 | 35  | 1    |

```
1 nrow(d)
```

```
[1] 544
```

```
1 nrow(d2)
```

```
[1] 352
```



# How to bake a Bayesian model

- . *Define the likelihood*
- . Set priors
- . Check that our priors are reasonable with simulation
- . Condition on the data
- . Visualize the posterior

## Defining our likelihood

$$h_i \sim \text{Normal}(\mu, \sigma)$$

# How to bake a Bayesian model

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## Simulate 4 priors for $\mu$

```
1 all_the_priors <- data.frame(  
2   sigma1 = rnorm(1e4, 178, 1),  
3   sigma2 = rnorm(1e4, 178, 2),  
4   sigma10 = rnorm(1e4, 178, 10),  
5   sigma25 = rnorm(1e4, 178, 20))
```

# Distributions of priors for $\mu$

```
1 all_the_priors |>
2   pivot_longer(cols = everything()) |>
3   ggplot(aes(x = value, color = name)) +
4   geom_density() +
5   labs(x = "Height in cm", color = "Mu prior")
```

---

## Now for $\sigma$

- How much do individuals vary on average from the population average? How variable is height?
- Recall  $sd = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$
- The average deviation of an observation from the mean
- $\sigma$  is constrained to be positive, our prior should reflect this

## The impact of $\sigma$ on our prior

Our prior for  $\sigma$  captures our beliefs about how much variance there is within the population.

*How much variable is height across adults?*

- No idea how variable the population is:  $\sigma \sim \text{Uniform}(0, 50)$
- We know there's not much variation:  $\sigma \sim \text{Uniform}(0, 10)$

# How to bake a Bayesian model

- . Define the likelihood
- . Set priors
- . *Check that our priors are reasonable with simulation*
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# Prior simulation

## . Simulate $\mu$

```
1 ### simulate mu prior: mu ~ N(178, 20)
2 mu_sims<-rnorm(1e4, 178, 20)
```

## . Simulate $\sigma$

```
1 ### simulate sigma prior: sigma ~ U(0,50)
2 sigma_sims<-runif(1e5, 0, 50)
```

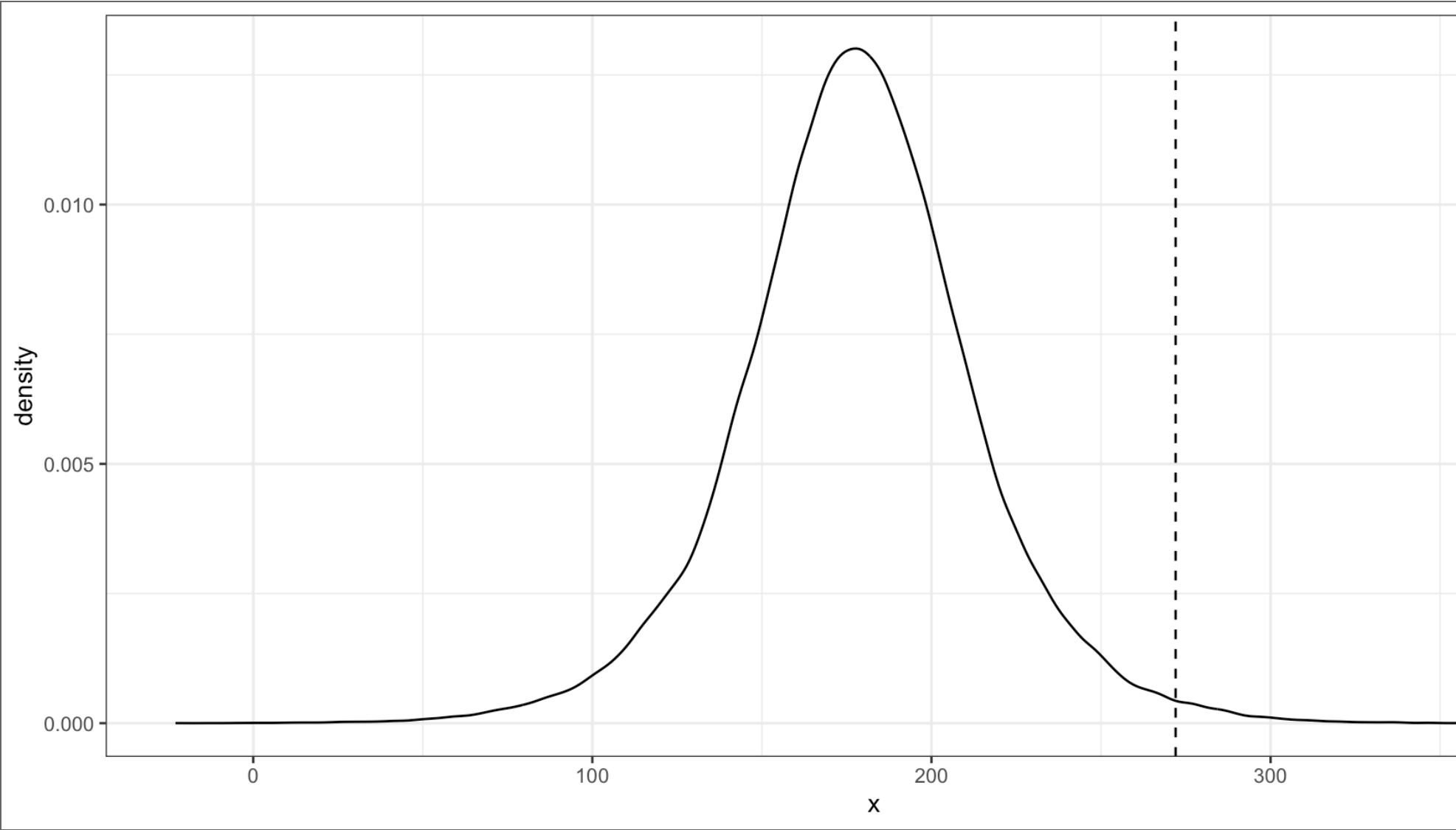
## . Draw predictions from $N(\mu, \sigma)$

```
1 ### prior predictive distribution: y~N(mu, sigma)
2 height_preds<-rnorm(1e5, mu_sims, sigma_sims)
```

# Visualize our prior

---

# Visualize prior predictions





# Fitting the model

# Recipe

- . Define the likelihood
- . Set priors
- . Check that our priors are reasonable with simulation
- . *Condition on the data*
- . Visualize the posterior

## Math -> R (quap)

Likelihood:  $h_i \sim \text{Normal}(\mu, \sigma)$

$\mu$  Prior:  $\mu \sim \text{Normal}(178, 20)$

$\sigma$  Prior:  $\sigma \sim \text{Uniform}(0, 50)$

```
1 ## define model
2 model_formula<-alist(
3   height ~ dnorm(mu, sigma),
4   mu ~ dnorm(178, 20),
5   sigma ~ dunif(0,50)
6 )
7 ## estimate with quap
8 m0<-quap(model_formula, data = d2)
9 summary(m0)
```

|       | mean       | sd        | 5.5%       | 94.5%      |
|-------|------------|-----------|------------|------------|
| mu    | 154.607240 | 0.4119208 | 153.948911 | 155.265569 |
| sigma | 7.729945   | 0.2912553 | 7.264463   | 8.195427   |

## Math -> R (brms)

```
1 prior_mu <- prior(normal(178,20), class = "Intercept")
2 prior_sigma <- prior(uniform(0, 50), class = "sigma", ub=50)
3 priors <- c(prior_mu, prior_sigma)
4 b0 <- brm(height ~ 1,
5           family = gaussian(),
6           prior = priors,
7           data = d2,
8           silent = 2)
```

SAMPLING FOR MODEL 'anon\_model' NOW (CHAIN 1).

Chain 1:

Chain 1: Gradient evaluation took 3e-05 seconds

Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.3 seconds.

Chain 1: Adjust your expectations accordingly!

Chain 1:

Chain 1:

Chain 1: Iteration: 1 / 2000 [ 0%] (Warmup)

Chain 1: Iteration: 200 / 2000 [ 10%] (Warmup)

Chain 1: Iteration: 400 / 2000 [ 20%] (Warmup)

Chain 1: Iteration: 600 / 2000 [ 30%] (Warmup)

Chain 1: Iteration: 800 / 2000 [ 40%] (Warmup)

Chain 1: Iteration: 1000 / 2000 [ 50%] (Warmup)

Chain 1: Iteration: 1001 / 2000 [ 50%] (Sampling)

Chain 1: Iteration: 1200 / 2000 [ 60%] (Sampling)

Chain 1: Iteration: 1400 / 2000 [ 70%] (Sampling)

Chain 1: Iteration: 1600 / 2000 [ 80%] (Sampling)

Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)

Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)

Chain 1:

Chain 1: Elapsed Time: 0.021 seconds (Warm-up)

# Model output

```
1 summary(m0)
```

|       | mean       | sd        | 5.5%       | 94.5%      |
|-------|------------|-----------|------------|------------|
| mu    | 154.607240 | 0.4119208 | 153.948911 | 155.265569 |
| sigma | 7.729945   | 0.2912553 | 7.264463   | 8.195427   |

# Model output

```
1 summary(b0)
```

```
Family: gaussian
Links: mu = identity
Formula: height ~ 1
Data: d2 (Number of observations: 352)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

|           | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-----------|----------|-----------|----------|----------|------|----------|----------|
| Intercept | 154.61   | 0.41      | 153.80   | 155.43   | 1.00 | 3836     | 2575     |

Further Distributional Parameters:

|       | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------|----------|-----------|----------|----------|------|----------|----------|
| sigma | 7.77     | 0.30      | 7.21     | 8.39     | 1.00 | 3585     | 2477     |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

# Recipe

- . Define the likelihood
- . Set priors
- . Check that our priors are reasonable with simulation
- . Condition on the data
- . *Visualize the posterior*

# Sample the posterior: rethinking

```
1 post_m0 <- extract.samples(m0)
2 head(post_m0)
```

|   | mu       | sigma    |
|---|----------|----------|
| 1 | 155.0864 | 7.588096 |
| 2 | 154.6657 | 7.536422 |
| 3 | 154.5440 | 7.672469 |
| 4 | 154.2215 | 7.626692 |
| 5 | 154.0791 | 7.554547 |
| 6 | 154.7865 | 8.220775 |

# Sample the posterior: tidybayes + brms

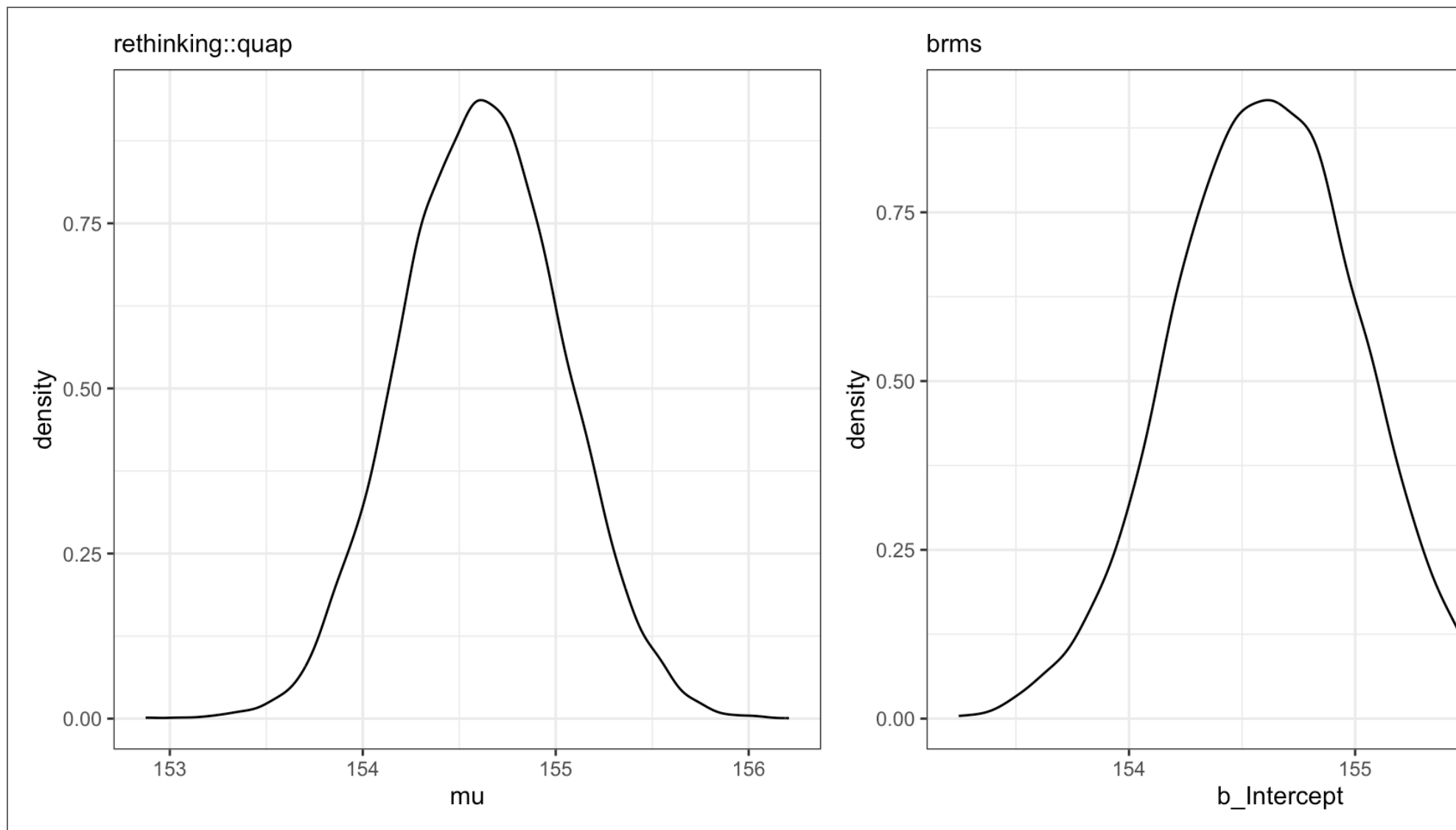
```
1 post_b0 <- as_draws_df(b0)
2 head(post_b0)
```

```
# A draws_df: 6 iterations, 1 chains, and 5 variables
```

|   | b_Intercept | sigma | Intercept | lprior | lp__  |
|---|-------------|-------|-----------|--------|-------|
| 1 | 155         | 7.5   | 155       | -8.5   | -1227 |
| 2 | 155         | 7.8   | 155       | -8.5   | -1226 |
| 3 | 155         | 8.0   | 155       | -8.5   | -1227 |
| 4 | 155         | 8.0   | 155       | -8.5   | -1227 |
| 5 | 155         | 7.9   | 155       | -8.5   | -1226 |
| 6 | 154         | 7.7   | 154       | -8.5   | -1227 |

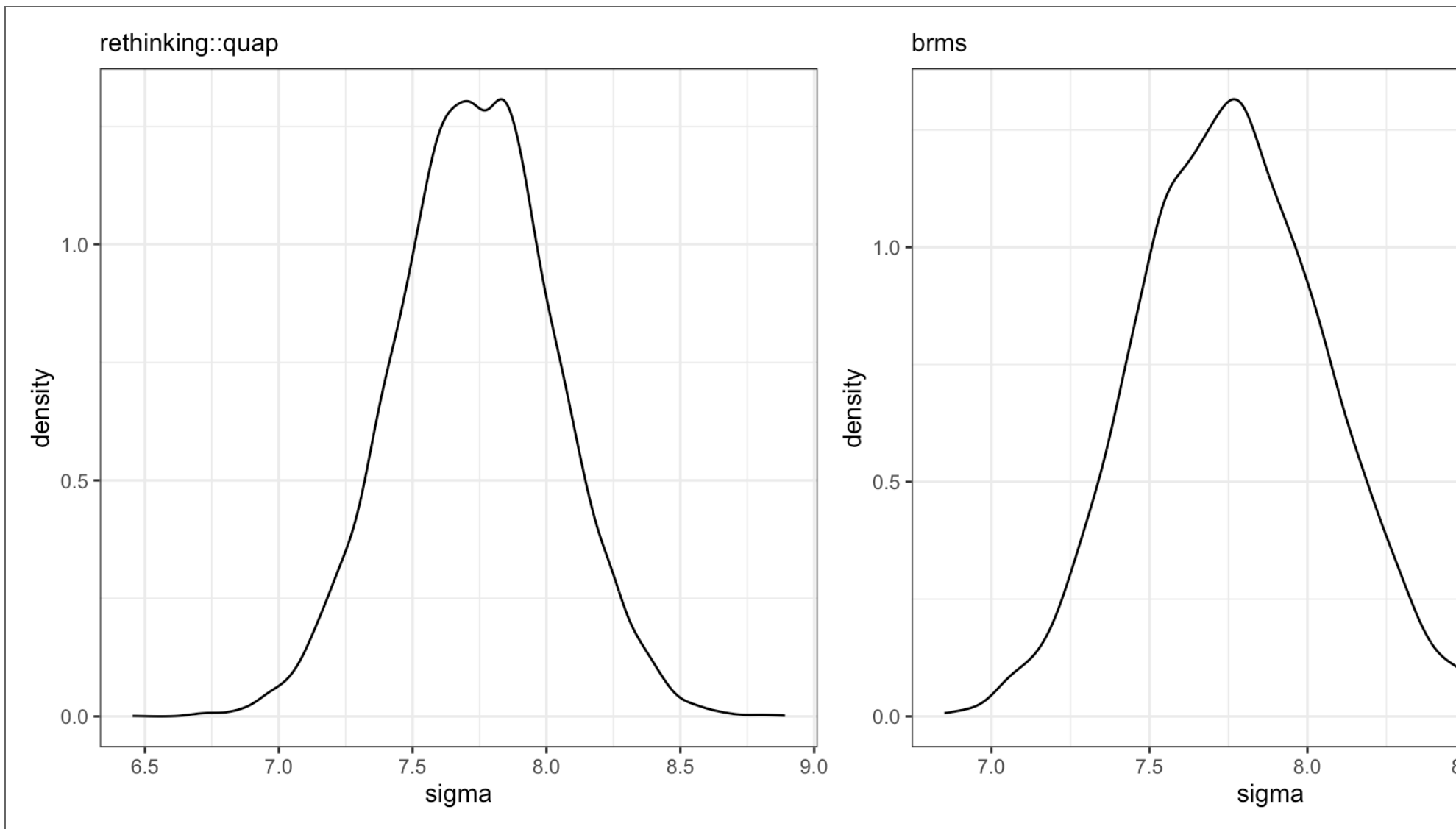
```
# ... hidden reserved variables {'.chain', '.iteration', '.draw'}
```

## Summarizing the posterior: mu





# Summarizing the posterior: sigma

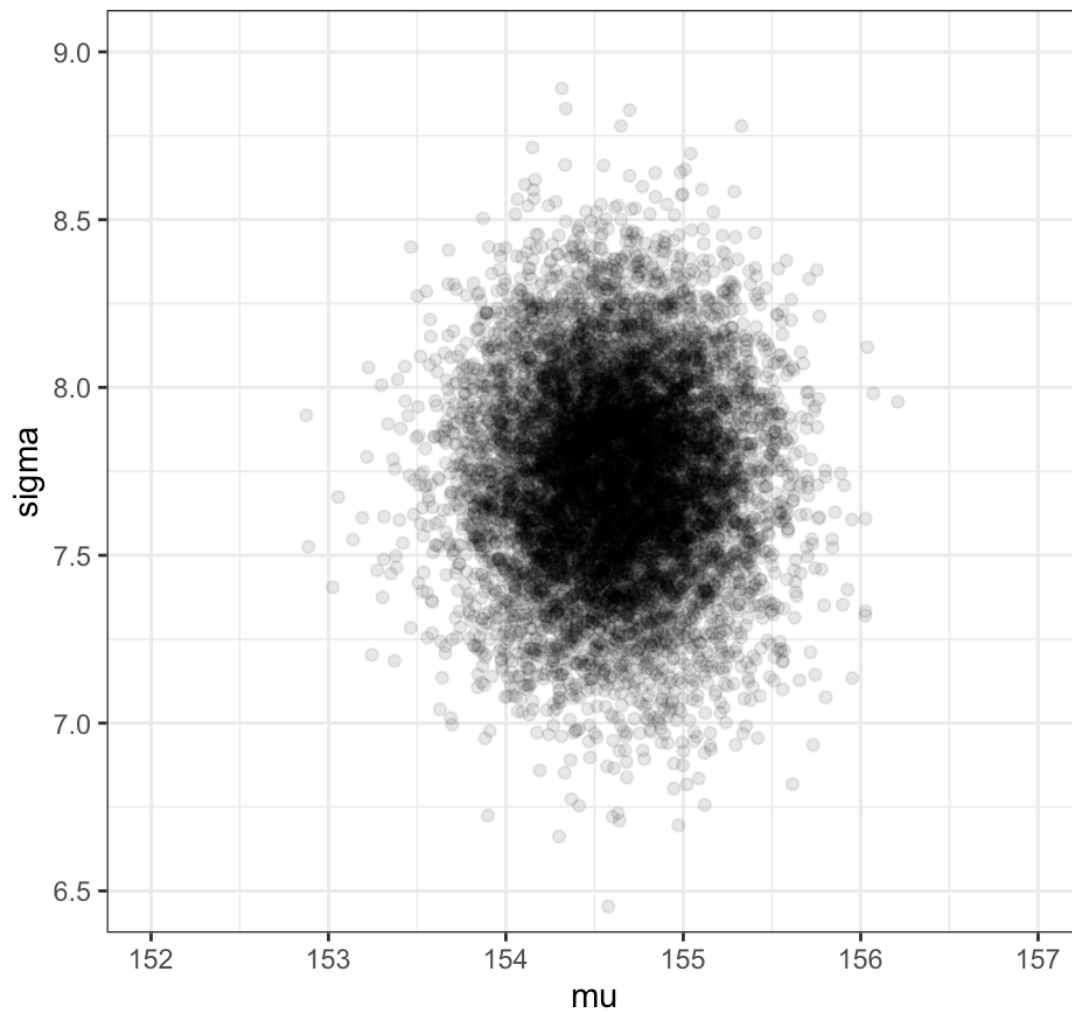




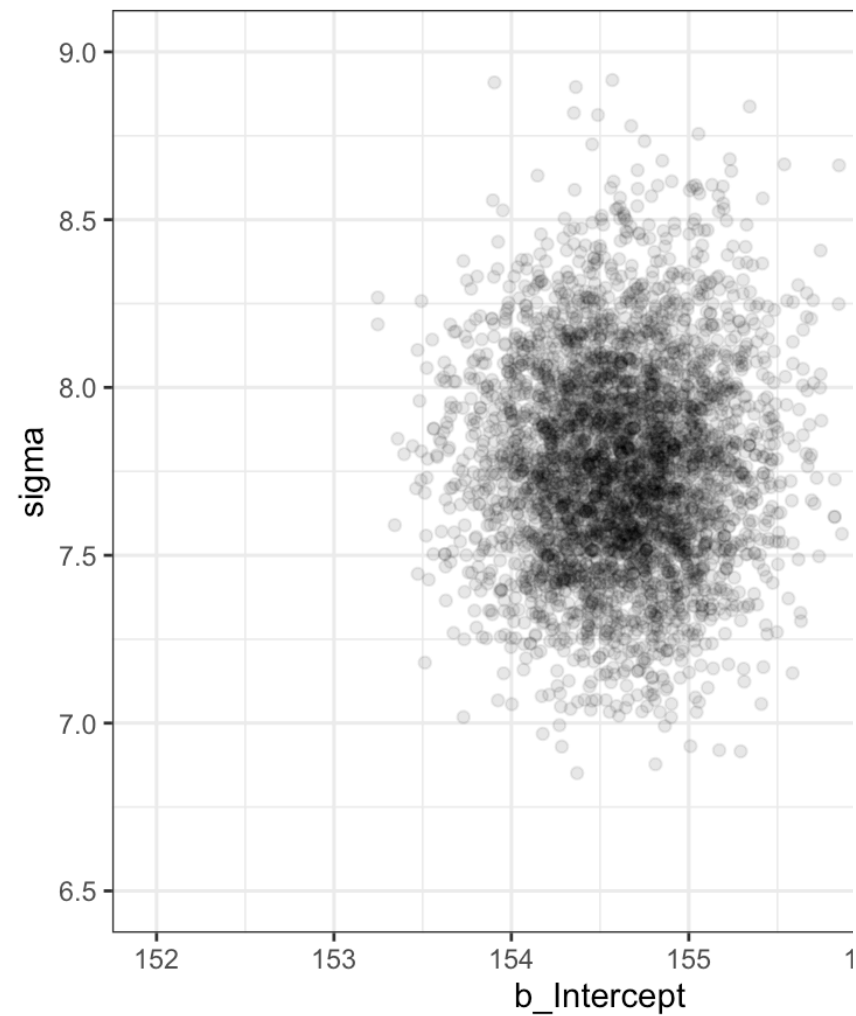
# The full posterior

```
1 p1 <- ggplot(post_m0, aes(x = mu, y = sigma)) +  
2   geom_point(alpha = 0.1) +  
3   labs(subtitle = "rethinking::quap") +  
4   coord_cartesian(xlim=c(152, 157), ylim = c(6.5, 9))  
5 p2 <- ggplot(post_b0, aes(x = b_Intercept, y = sigma)) +  
6   geom_point(alpha = 0.1) +  
7   labs(subtitle = "brms") +  
8   coord_cartesian(xlim=c(152, 157), ylim = c(6.5, 9))  
9 p1+p2
```

rethinking::quap



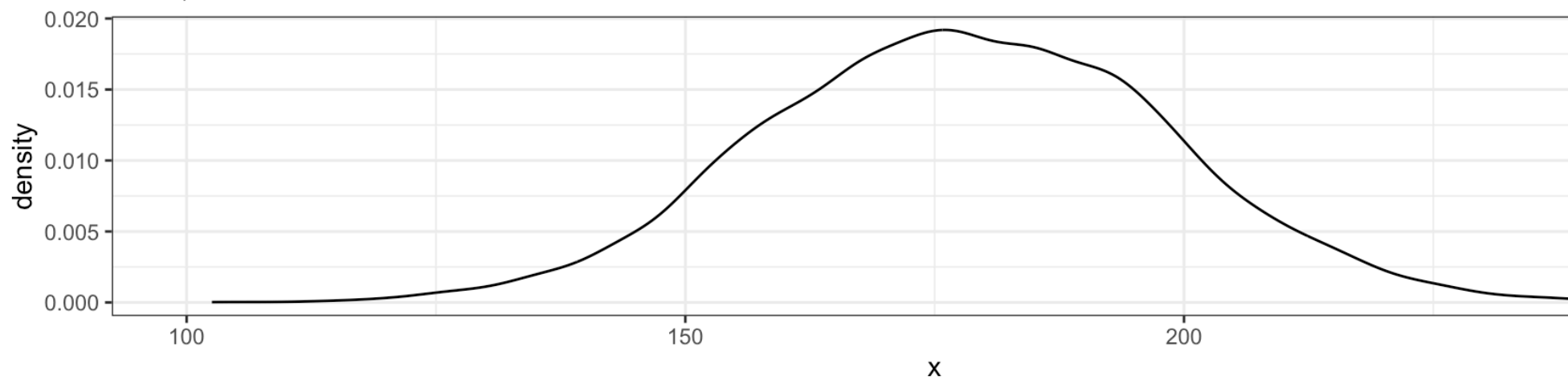
brms



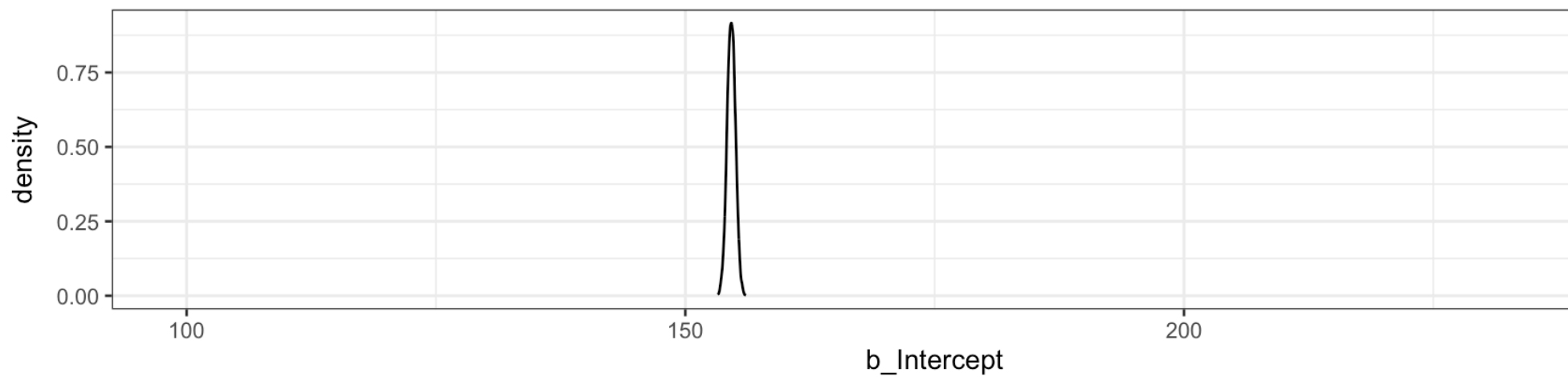
# Compare results to the prior

```
1 p1<-data.frame(x = mu_sims) |>
2   ggplot(aes(x = x)) +
3   geom_density() +
4   ggtitle("Prior, mu")+
5   coord_cartesian(xlim=c(100, 250))
6
7 p2<-ggplot(post_b0,
8             aes(x = b_Intercept)) +
9   geom_density() +
10  ggtitle("Posterior, mu") +
11  coord_cartesian(xlim=c(100, 250))
12
13 p1/p2
```

Prior,  $\mu$



Posterior,  $\mu$



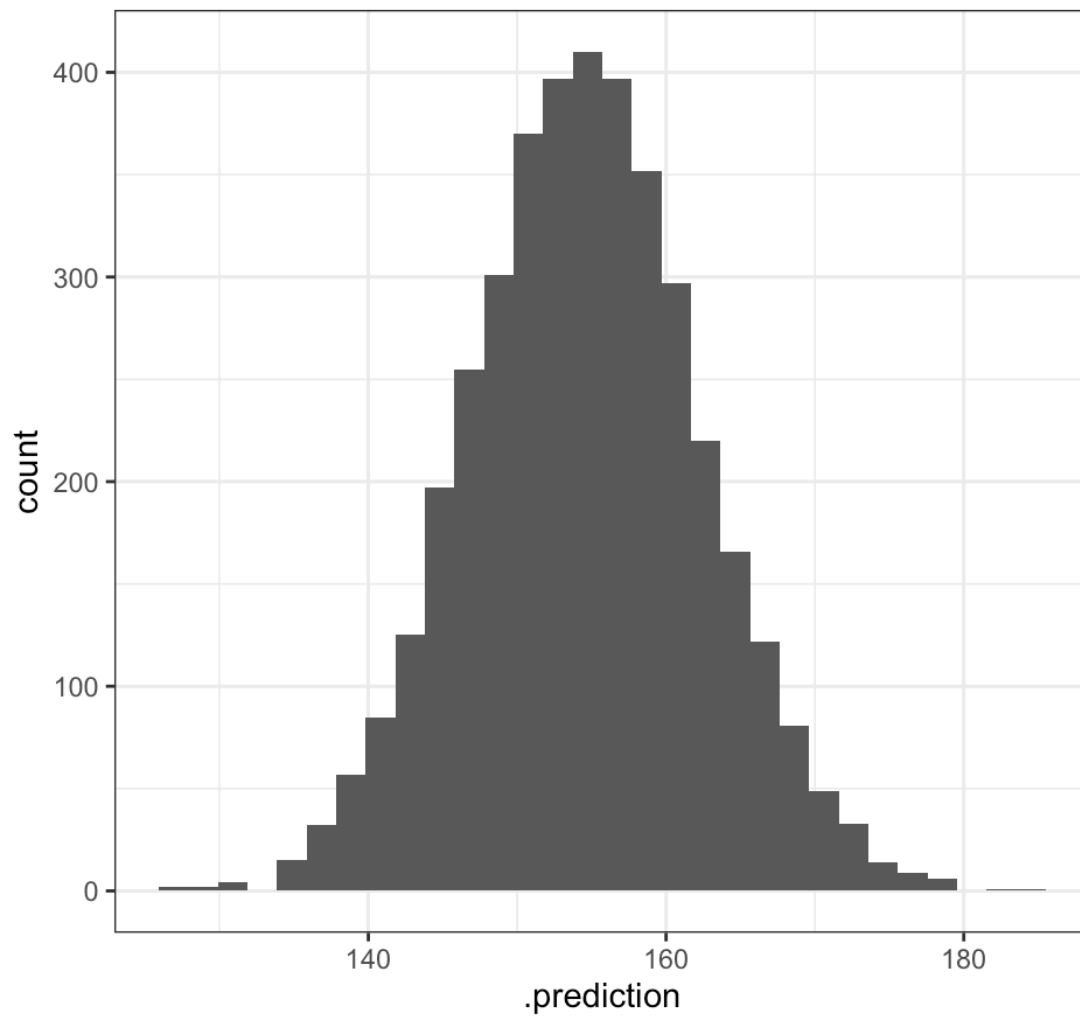
# Posterior prediction

```
1 # rethinking
2 post_pred1 <- rnorm(1e4, post_m0$mu, post_m0$sigma)
3 # brms manual mode
4 post_pred2 <- rnorm(1e4, post_b0$b_Intercept, post_b0$sigma)
5 # brms easy mode, just take the first row for 'newdata' for now
6 post_pred3 <- predicted_draws(b0, d2[1,])
```

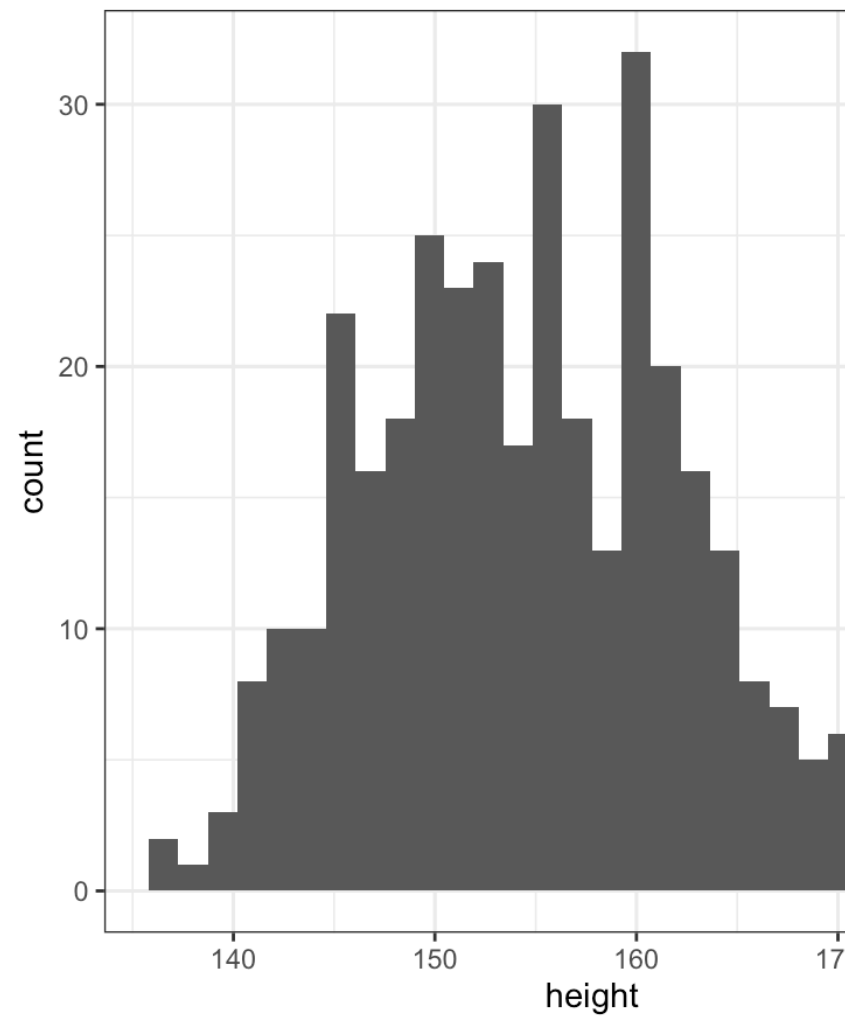
# Visualize posterior predictions

```
1 post_pred <- ggplot(post_pred3,  
2   aes(x = .prediction)) +  
3   geom_histogram() +  
4   labs(subtitle = "Posterior p[redictions"]  
5 observed <- ggplot(d2,  
6   aes(x = height)) +  
7   geom_histogram() +  
8   labs(subtitle = "Observed")  
9 post_pred+observed
```

Posterior p[redictions]

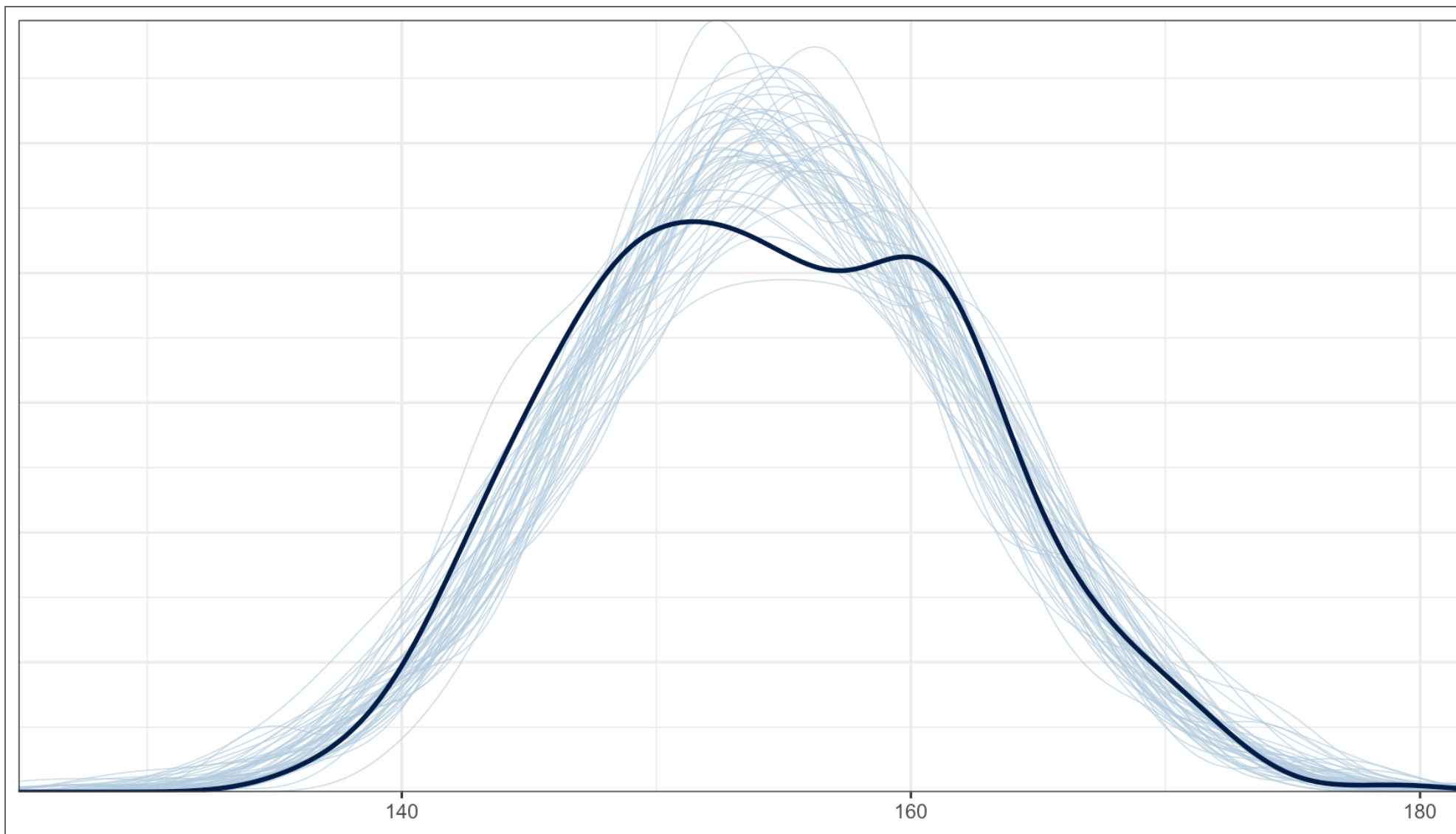


Observed



# Visualize posterior predictions (easy mode)

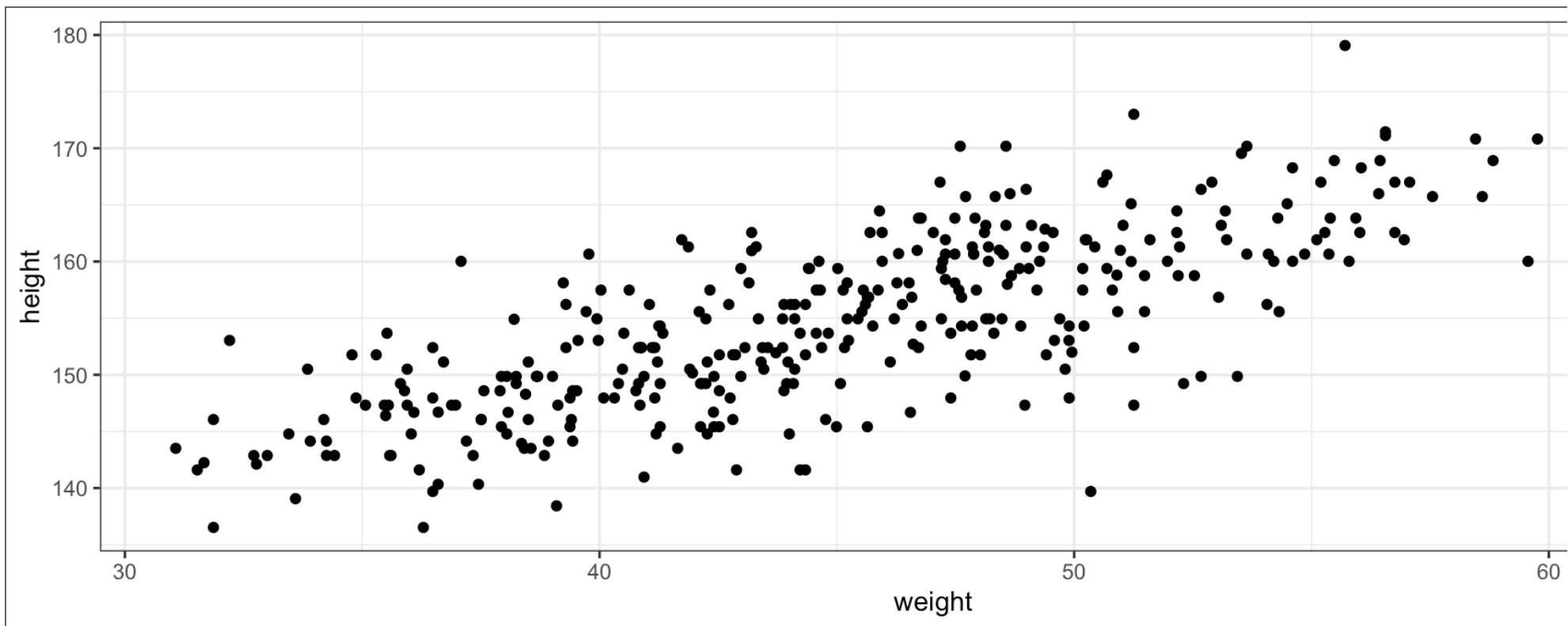
```
1 pp_check(b0, ndraws = 50)
```



# Moving beyond an intercept-only model

# Predicting height with other variables

```
1 ggplot(d2,  
2       aes(x = weight, y = height)) +  
3       geom_point()
```



# Recipe

- . *Define the likelihood*
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# The anatomy of a linear model

Begin with the Normal model of height

$$h_i \sim \text{Normal}(\mu, \sigma)$$

$$\mu \sim \text{Normal}(178, 20)$$

$$\sigma \sim \text{Uniform}(0, 50)$$

## Changing how we define mu

Let  $x_i$  be the weight of the person in row  $i$

$$h_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta(x_i - \bar{x})$$

# Recipe

- . Define the likelihood
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- . Check that our priors are reasonable with simulation
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## Priors for our new parameters

Our prior now describes a line:  $\mu = \alpha + \beta x$

$\alpha \sim \text{Normal}(178, 20)$

$\beta \sim \text{Normal}(0, 10)$

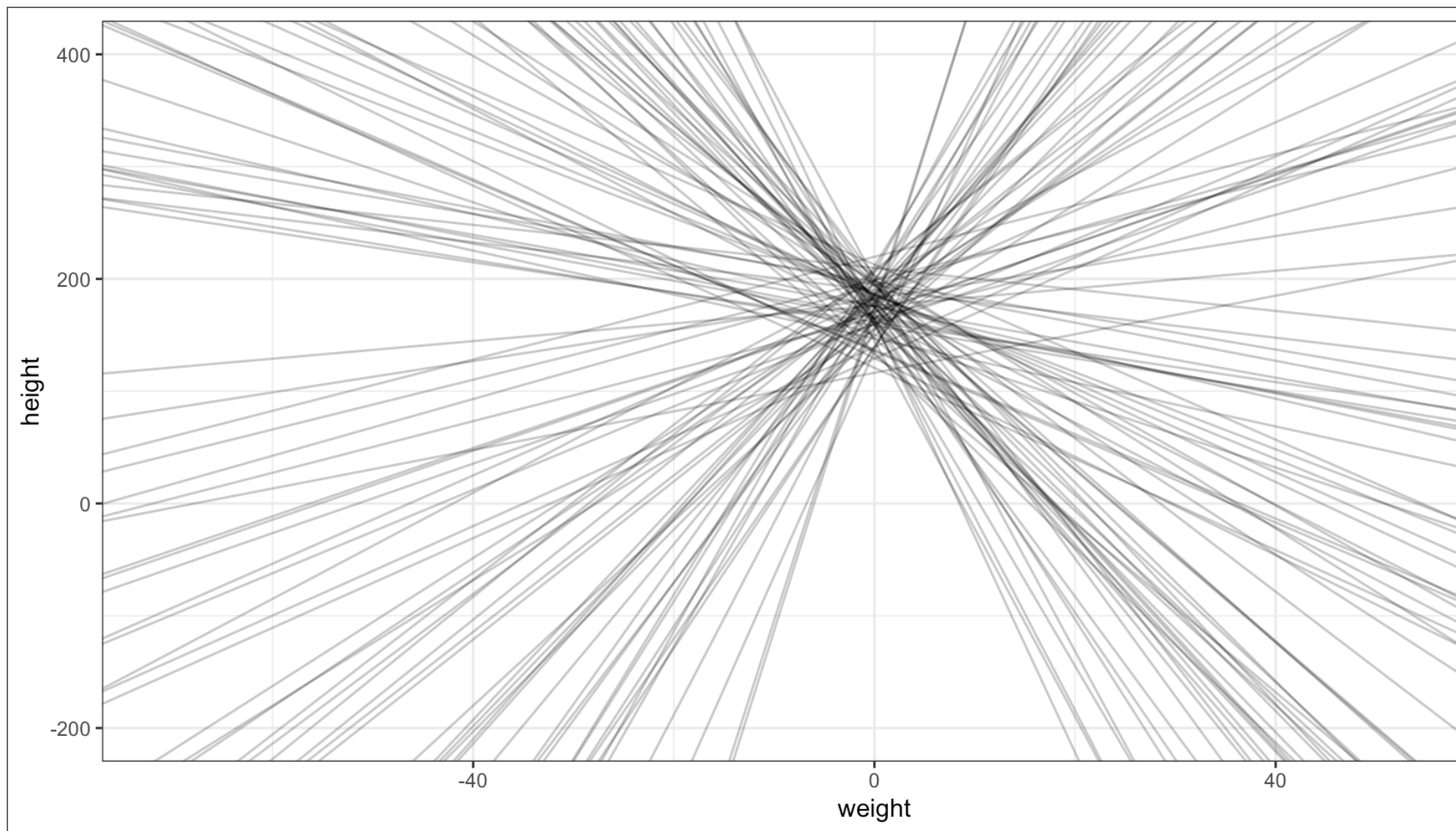
# Recipe

- . Define the likelihood
- . Set priors
- . *Check that our priors are reasonable with simulation*
- . Condition on the data
- . Visualize the posterior

# What does our prior imply?

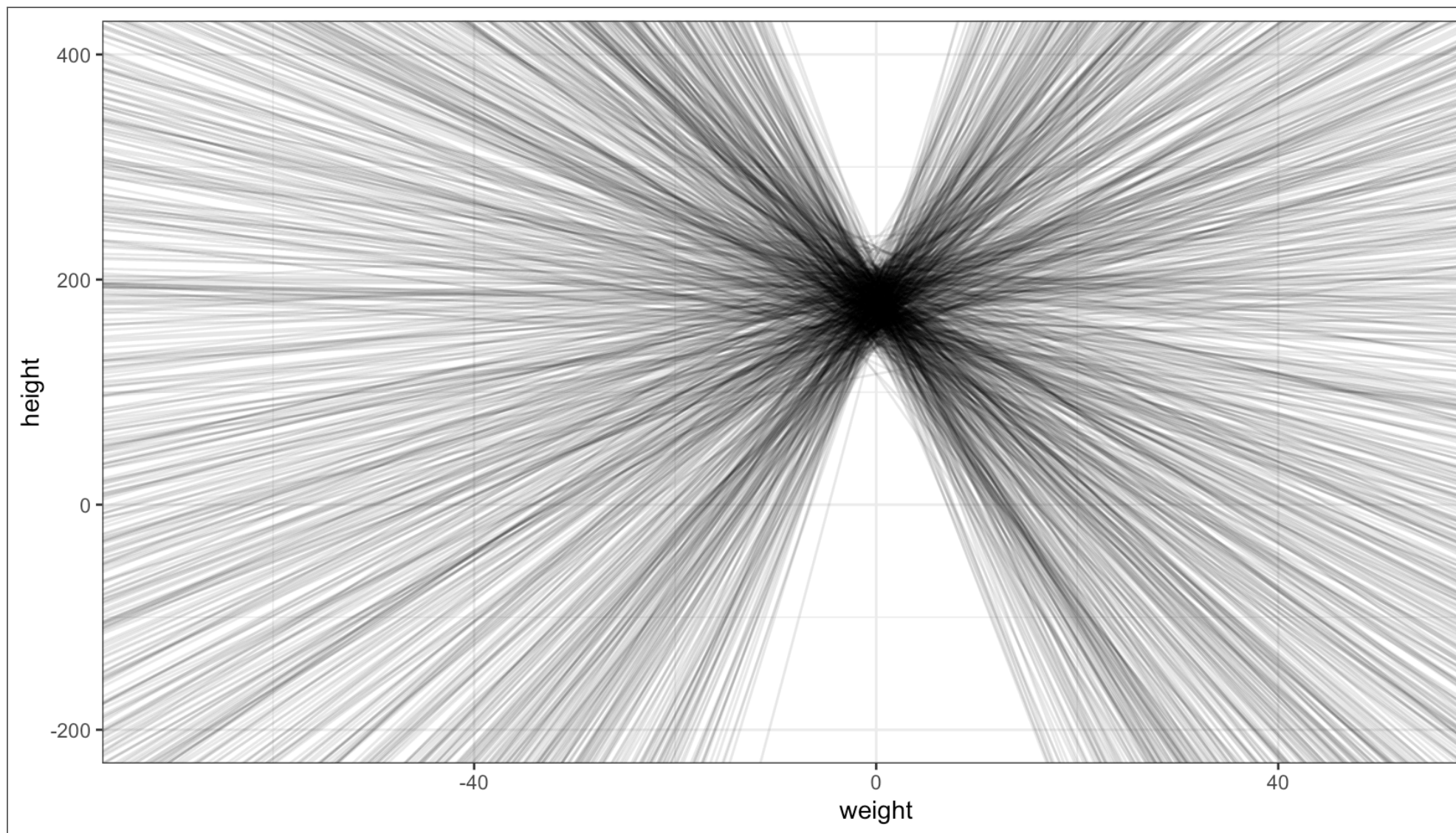
```
1 ## Simulate regression lines from prior and visualize
2 n<-100
3 alpha<-rnorm(n, 178, 20)
4 beta<-rnorm(n, 0, 10)
```

# What does our prior imply?





# What does our prior imply?

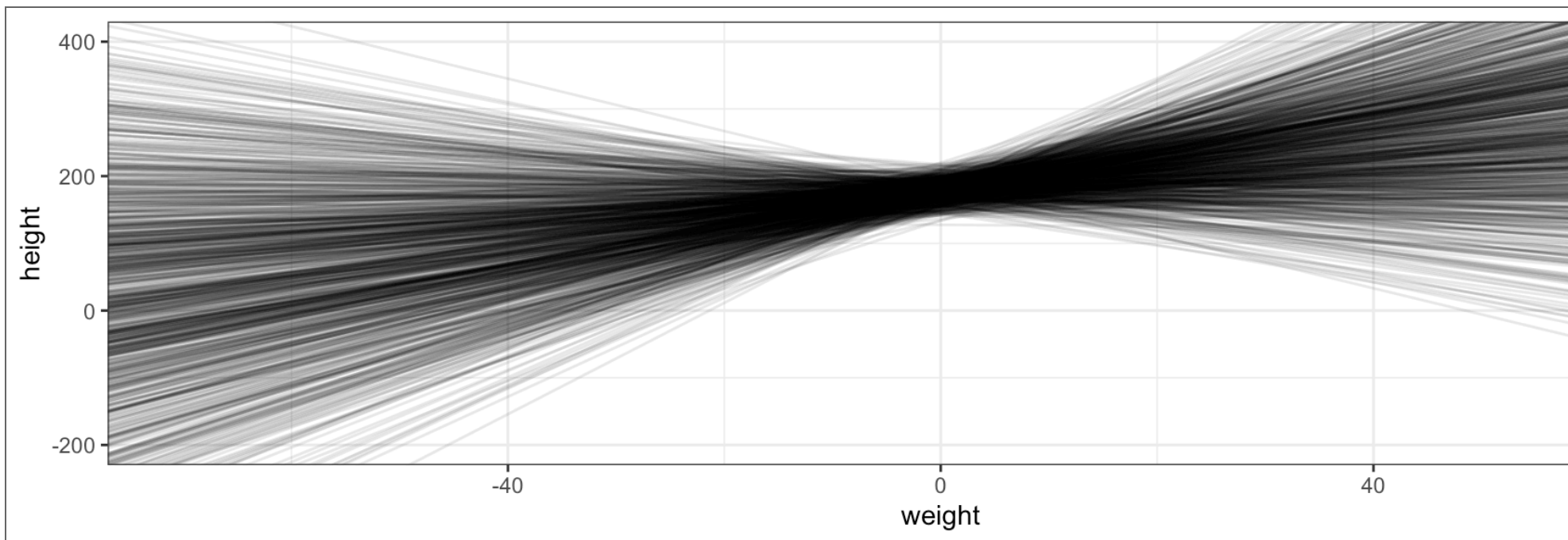




## Let's try again

Let's shrink the spread on  $\alpha$ , and set  $\beta$  to be more likely positive

```
1 ## Simulate regression lines from prior and visualize
2 n<-1000
3 alpha<-rnorm(n, 178, 15)
4 beta<-rnorm(n, 2, 2)
5 plot_dat<-data.frame(alpha = alpha, beta=beta, sim_n = factor(1:n),
6                       weight = seq(-70, 70, length.out=n),
7                       height = seq(-200,400, length.out=n))
8 ggplot(plot_dat,
9         aes(x = weight, y = height)) +
10   geom_blank() +
11   geom_abline(aes(intercept = alpha,
12                   slope = beta,
13                   group= sim_n),
14               alpha = 0.1)
```



# Recipe

- . Define the likelihood
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## Estimating the model

- Likelihood for height:  $h_i \sim \text{Normal}(\mu_i, \sigma)$
- Linear model for mean:  $\mu_i = \alpha + \beta x_i$
- Prior for intercept:  $\alpha \sim \text{Normal}(178, 15)$
- Prior for slope:  $\beta \sim \text{Normal}(2, 2)$
- Prior for standard deviation:  $\sigma \sim \text{Uniform}(0, 50)$

# Estimation

```
1 # create x-xbar centered weight
2 d2 <- d2 |>
3   mutate(weight.c = weight - mean(weight))
4
5 prior_alpha <- prior(normal(178,15), class = "Intercept")
6 prior_beta <- prior(normal(2, 2), class = "b")
7 prior_sigma <- prior(uniform(0, 50), class = "sigma", ub=50)
8 priors <- c(prior_alpha, prior_beta, prior_sigma)
9 m1 <- brm(height ~ weight.c,
10          family = gaussian(),
11          prior = priors,
12          data = d2,
13          silent = 2)
```

SAMPLING FOR MODEL 'anon\_model' NOW (CHAIN 1).

Chain 1:

Chain 1: Gradient evaluation took 2.3e-05 seconds

Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.23 seconds.

Chain 1: Adjust your expectations accordingly!

Chain 1:

Chain 1:

Chain 1: Iteration: 1 / 2000 [ 0%] (Warmup)

Chain 1: Iteration: 200 / 2000 [ 10%] (Warmup)

Chain 1: Iteration: 400 / 2000 [ 20%] (Warmup)

Chain 1: Iteration: 600 / 2000 [ 30%] (Warmup)

Chain 1: Iteration: 800 / 2000 [ 40%] (Warmup)

Chain 1: Iteration: 1000 / 2000 [ 50%] (Warmup)

Chain 1: Iteration: 1001 / 2000 [ 50%] (Sampling)  
Chain 1: Iteration: 1200 / 2000 [ 60%] (Sampling)  
Chain 1: Iteration: 1400 / 2000 [ 70%] (Sampling)  
Chain 1: Iteration: 1600 / 2000 [ 80%] (Sampling)  
Chain 1: Iteration: 1800 / 2000 [ 90%] (Sampling)  
Chain 1: Iteration: 2000 / 2000 [100%] (Sampling)  
Chain 1:  
Chain 1: Elapsed Time: 0.006 seconds (Warm-up)

# Recipe

- . Define the likelihood
- . Set priors
- . Check that our priors are reasonable with simulation
- . Condition on the data
- . *Visualize the posterior*

# Interpreting the posterior: table

```
1 summary(m1)
```

```
Family: gaussian
Links: mu = identity
Formula: height ~ weight.c
Data: d2 (Number of observations: 352)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

|           | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-----------|----------|-----------|----------|----------|------|----------|----------|
| Intercept | 154.61   | 0.27      | 154.09   | 155.14   | 1.00 | 3847     | 3013     |
| weight.c  | 0.91     | 0.04      | 0.82     | 0.99     | 1.00 | 4049     | 2979     |

Further Distributional Parameters:

|       | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------|----------|-----------|----------|----------|------|----------|----------|
| sigma | 5.10     | 0.20      | 4.72     | 5.50     | 1.00 | 4114     | 3078     |

Draws were sampled using `sampling(NUTS)`. For each parameter, `Bulk_ESS` and `Tail_ESS` are effective sample size measures, and `Rhat` is the potential scale reduction factor on split chains (at convergence, `Rhat` = 1).

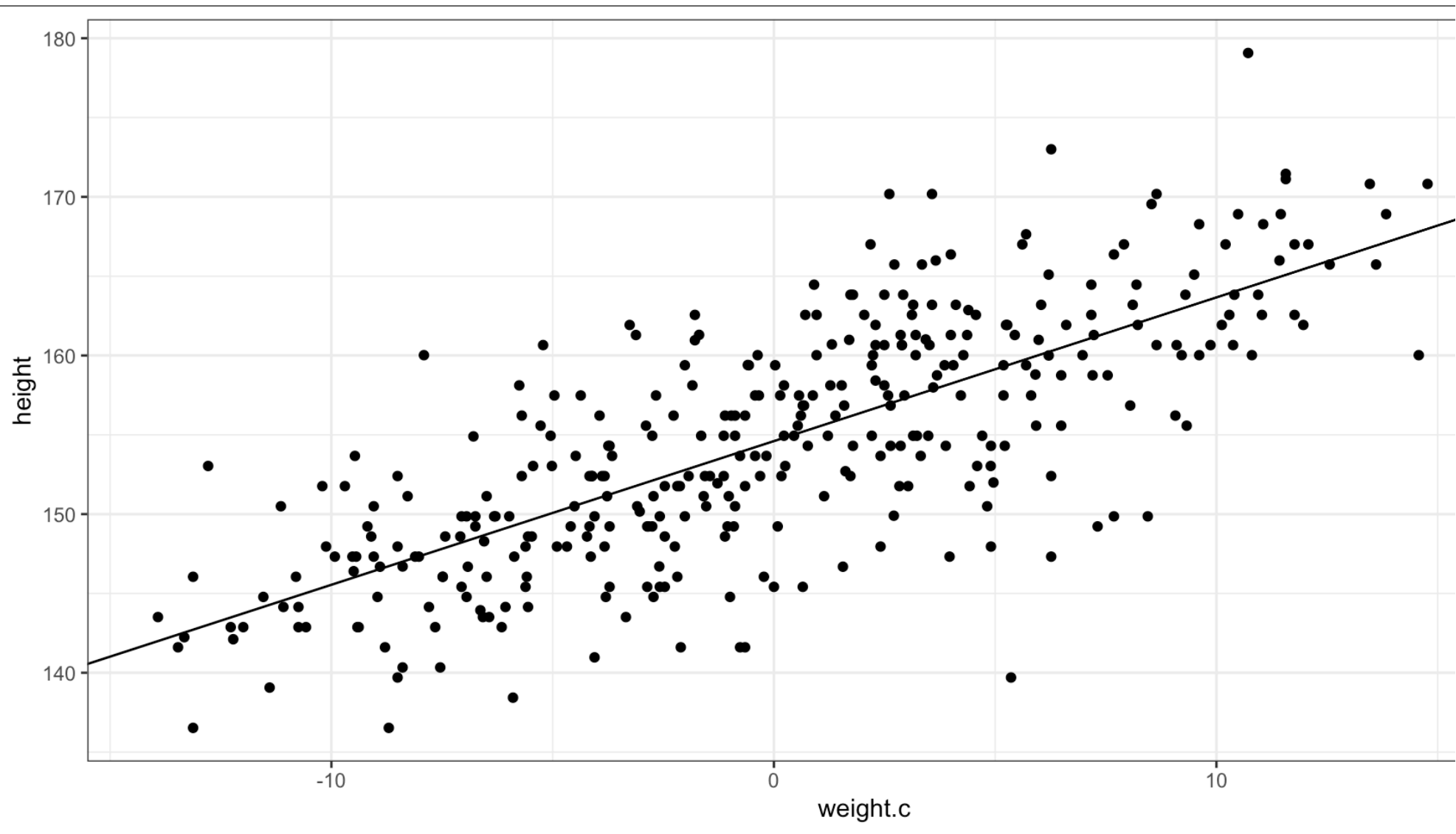
# Examining posterior samples

```
1 post_m1<-as_draws_df(m1) |>  
2   rename(alpha = b_Intercept, beta = b_weight.c) |>  
3   select(alpha, beta, sigma)  
4 head(post_m1)
```

```
# A tibble: 6 × 3  
  alpha  beta sigma  
  <dbl> <dbl> <dbl>  
1  155.  0.874  4.78  
2  154.  0.935  5.40  
3  155.  0.857  4.99  
4  154.  0.952  5.38  
5  155.  0.900  5.21  
6  155.  0.912  5.38
```

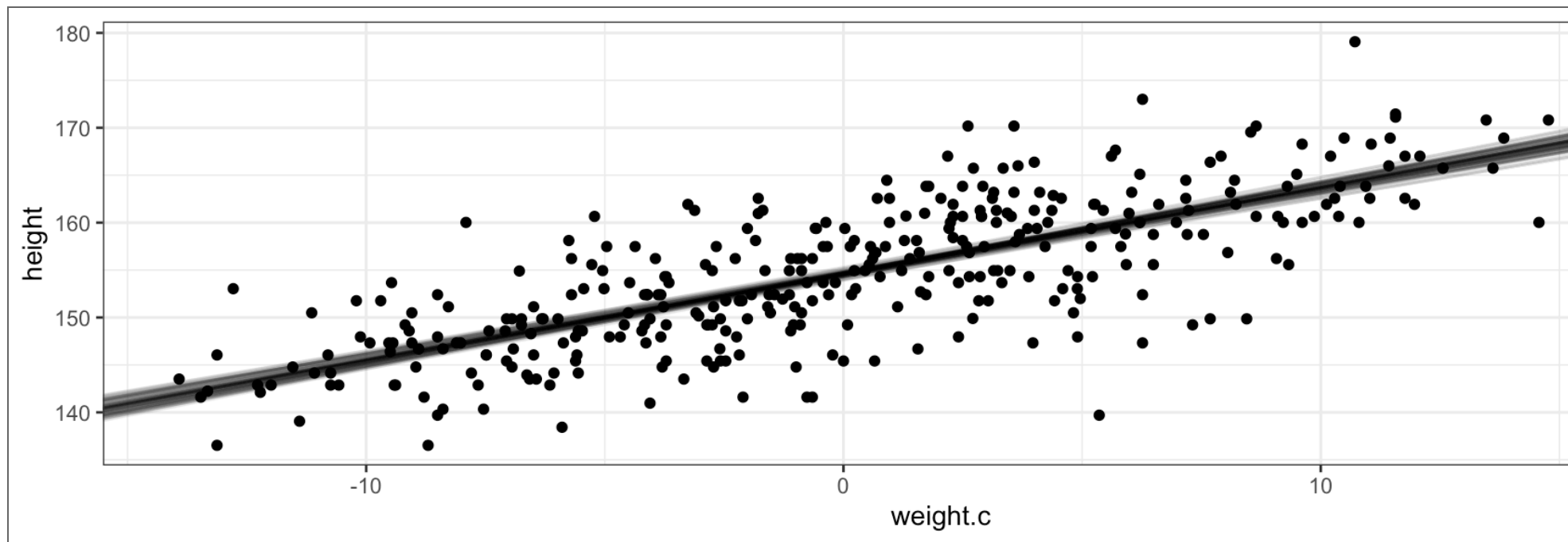
# Visualizing the estimated posterior: posterior mean

```
1 plot_dat<-post_m1 |>
2   summarise(a = mean(alpha), b = mean(beta))
3 ggplot(d2,
4   aes(x = weight.c, y = height)) +
5   geom_point() +
6   geom_abline(data = plot_dat,
7     aes(intercept = a, slope = b))
```



## But the posterior is more than one line

```
1 ggplot(d2,  
2       aes(x = weight.c, y = height)) +  
3   geom_point() +  
4   geom_abline(data = post_m1[1:100,],  
5             aes(intercept = alpha, slope = beta), alpha = 0.1)
```

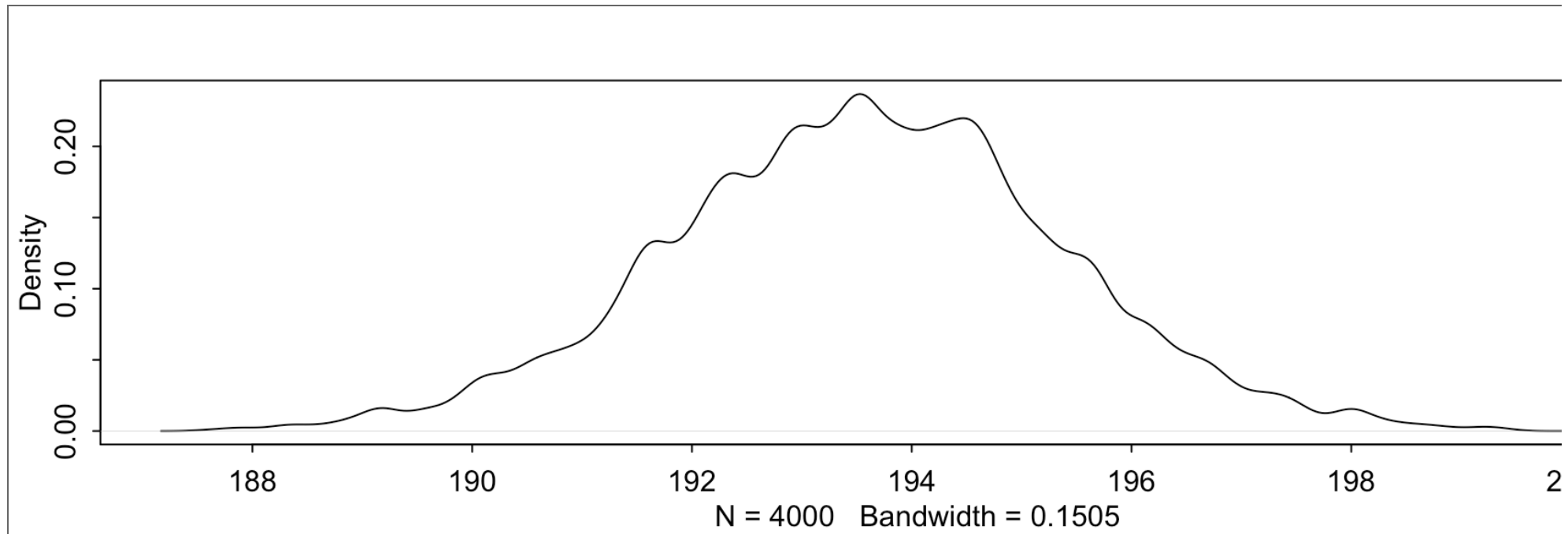


## At a single value of x

I weigh about 88 kg. Let's solve for  $h$

$$E(h_i) = \mu_i = \alpha + \beta(x_i - \bar{x})$$

```
1 mu_FE <- post_m1$alpha + post_m1$beta * (88 - mean(d2$weight))  
2 dens(mu_FE)
```

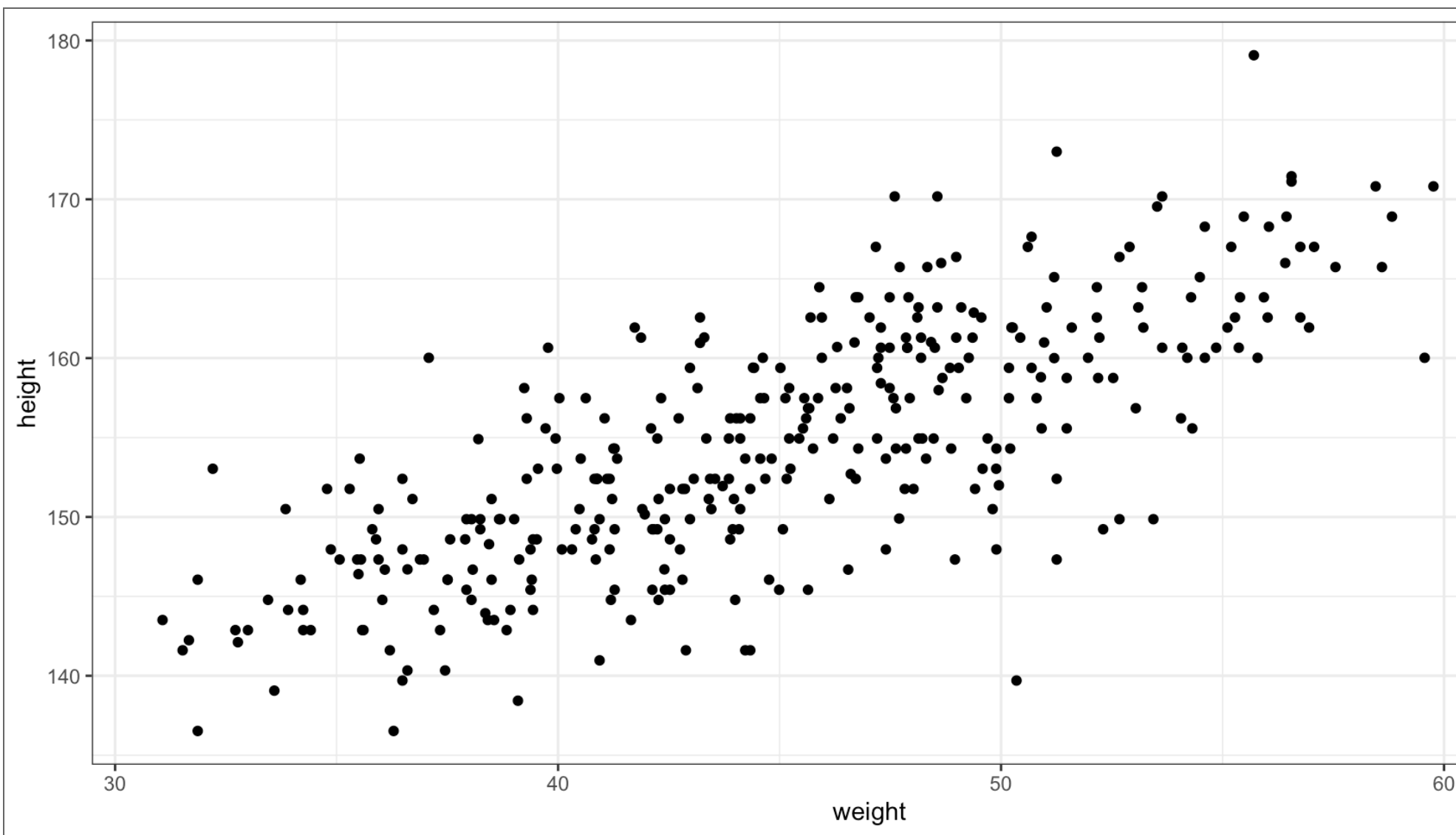


These are the values of  $\mu$  for individuals of weight 88kg that are compatible with the data and the model



# A problem: extrapolation!

```
1 ggplot(d2,  
2       aes(x = weight, y = height)) +  
3   geom_point()
```



# Summarizing the posterior over the full range of the data

```
1 ## create a vector for the sequence of values to predict
2 weights<-seq(min(d2$weight), max(d2$weight), by=1)
3 sim_dat<-data.frame(weight.c = weights - mean(d2$weight))
4 ## add_epred_draws adds posterior samples of the linear predictor for mu
5 mu<-sim_dat |>
6   add_epred_draws(m1)
7 head(mu)
```

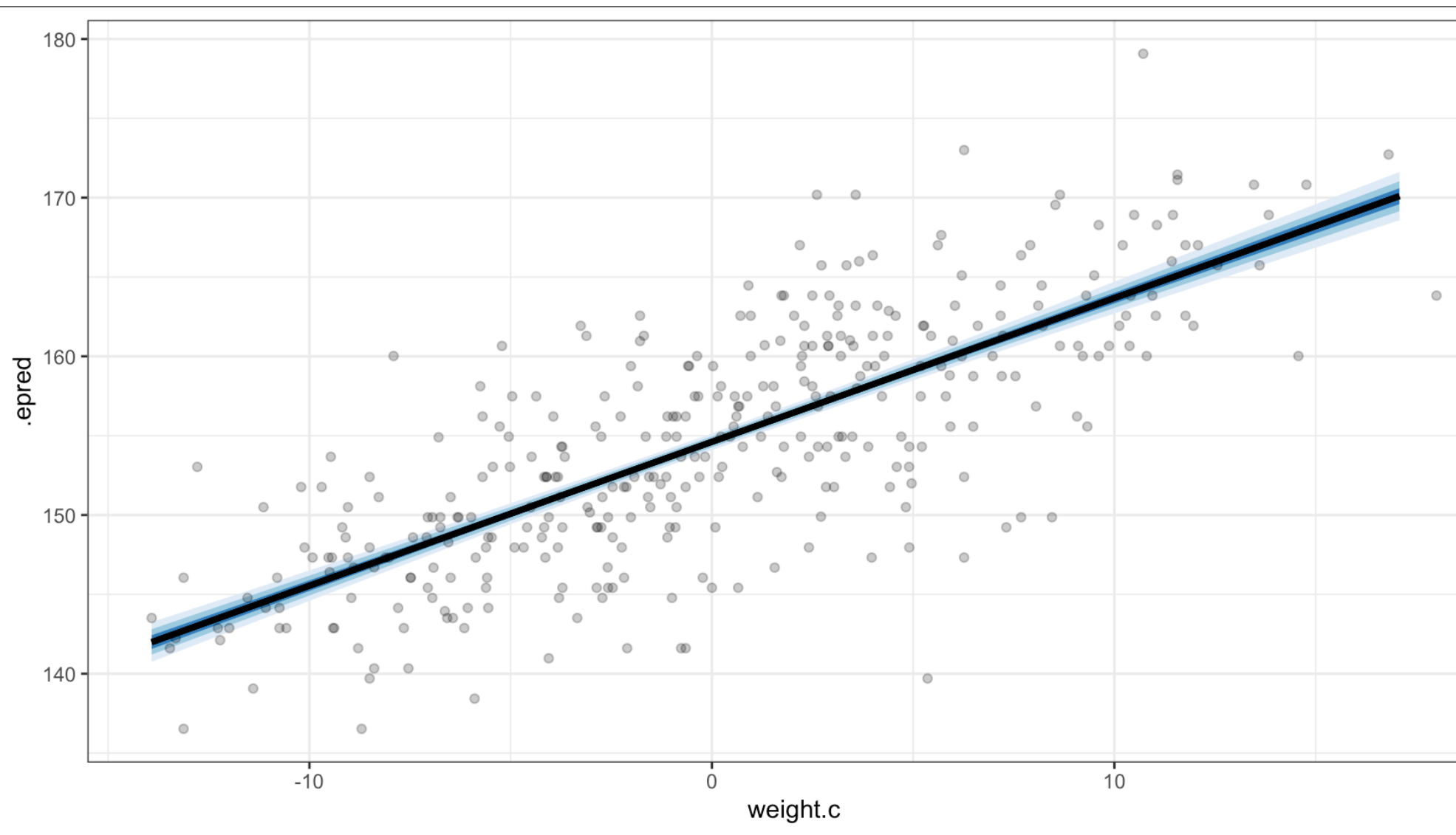
# A tibble: 6 × 6

# Groups: weight.c, .row [1]

|   | weight.c<br><dbl> | .row<br><int> | .chain<br><int> | .iteration<br><int> | .draw<br><int> | .epred<br><dbl> |
|---|-------------------|---------------|-----------------|---------------------|----------------|-----------------|
| 1 | -13.9             | 1             | NA              | NA                  | 1              | 143.            |
| 2 | -13.9             | 1             | NA              | NA                  | 2              | 141.            |
| 3 | -13.9             | 1             | NA              | NA                  | 3              | 143.            |
| 4 | -13.9             | 1             | NA              | NA                  | 4              | 141.            |
| 5 | -13.9             | 1             | NA              | NA                  | 5              | 142.            |
| 6 | -13.9             | 1             | NA              | NA                  | 6              | 142.            |

# Visualizing the posterior for mu: with CI

```
1 ggplot(mu,  
2       aes(x = weight.c, y = .epred)) +  
3   stat_lineribbon() +  
4   scale_fill_brewer(palette = "Blues") +  
5   geom_point(data = d2, aes(x = weight.c, y = height), alpha = 0.25)
```



# Predicting new data; validating the model

Recall the method for posterior prediction:

- . Draw samples of the parameters from the posterior
- . Use these samples to draw new predictions using the likelihood

```
1 ## already got the samples in post_m1
2 head(post_m1)
```

```
# A tibble: 6 × 3
  alpha  beta sigma
  <dbl> <dbl> <dbl>
1  155.  0.874  4.78
2  154.  0.935  5.40
3  155.  0.857  4.99
4  154.  0.952  5.38
5  155.  0.900  5.21
6  155.  0.912  5.38
```

We haven't used  $\sigma$  yet. It will tell our model how much sampling uncertainty to produce

# Method for posterior prediction (manual)

```
1 # iterate over all posterior uncertainty
2 n_sims <- nrow(post_m1)
3 # empty bucket for output
4 post_pred <- list()
5 for(i in 1:nrow(sim_dat)){
6   # set up deterministic component
7   linear_predictor <- post_m1$alpha + post_m1$beta * sim_dat$weight.c[i]
8   # and stochastic component, draw n_sims for each weight.c and posterior combination
9   post_pred[[i]] <- data.frame(ppd = rnorm(n_sims, mean = linear_predictor, sd = post_m1$sigma),
10                                weight.c = sim_dat$weight.c[i])
11 }
12 post_pred <- bind_rows(post_pred)
13 head(post_pred)
```

|   | ppd      | weight.c  |
|---|----------|-----------|
| 1 | 144.6760 | -13.91943 |
| 2 | 133.5283 | -13.91943 |
| 3 | 144.9353 | -13.91943 |
| 4 | 139.2127 | -13.91943 |
| 5 | 144.8912 | -13.91943 |
| 6 | 139.7916 | -13.91943 |

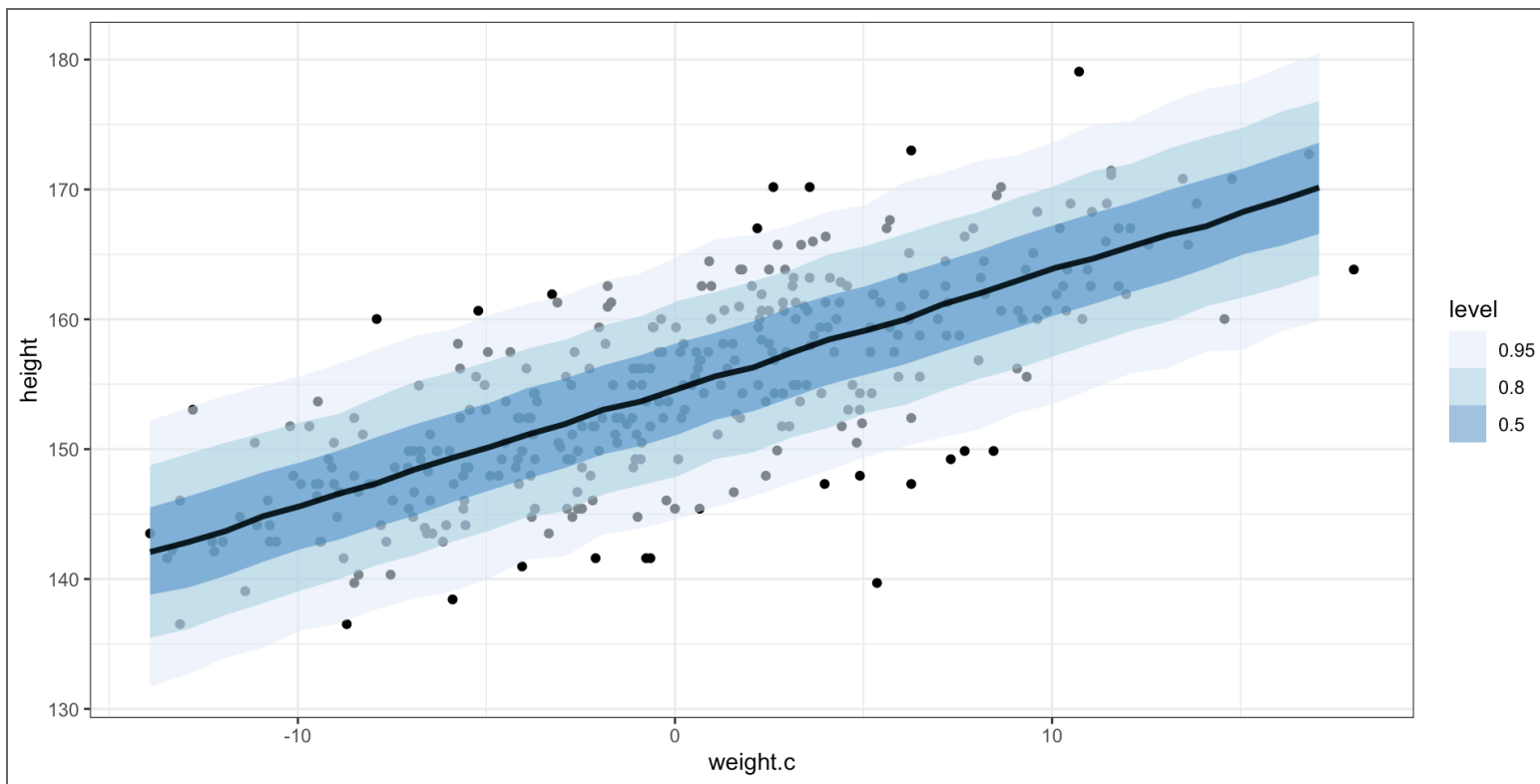
# Method for posterior prediction (easy mode)

```
1 sim_postpred <- sim_dat |>
2   add_predicted_draws(m1)
3 head(sim_postpred)
```

```
# A tibble: 6 × 6
# Groups:   weight.c, .row [1]
  weight.c .row .chain .iteration .draw .prediction
  <dbl> <int> <int> <int> <int> <dbl>
1 -13.9     1    NA      NA     1    149.
2 -13.9     1    NA      NA     2    140.
3 -13.9     1    NA      NA     3    145.
4 -13.9     1    NA      NA     4    140.
5 -13.9     1    NA      NA     5    137.
6 -13.9     1    NA      NA     6    142.
```

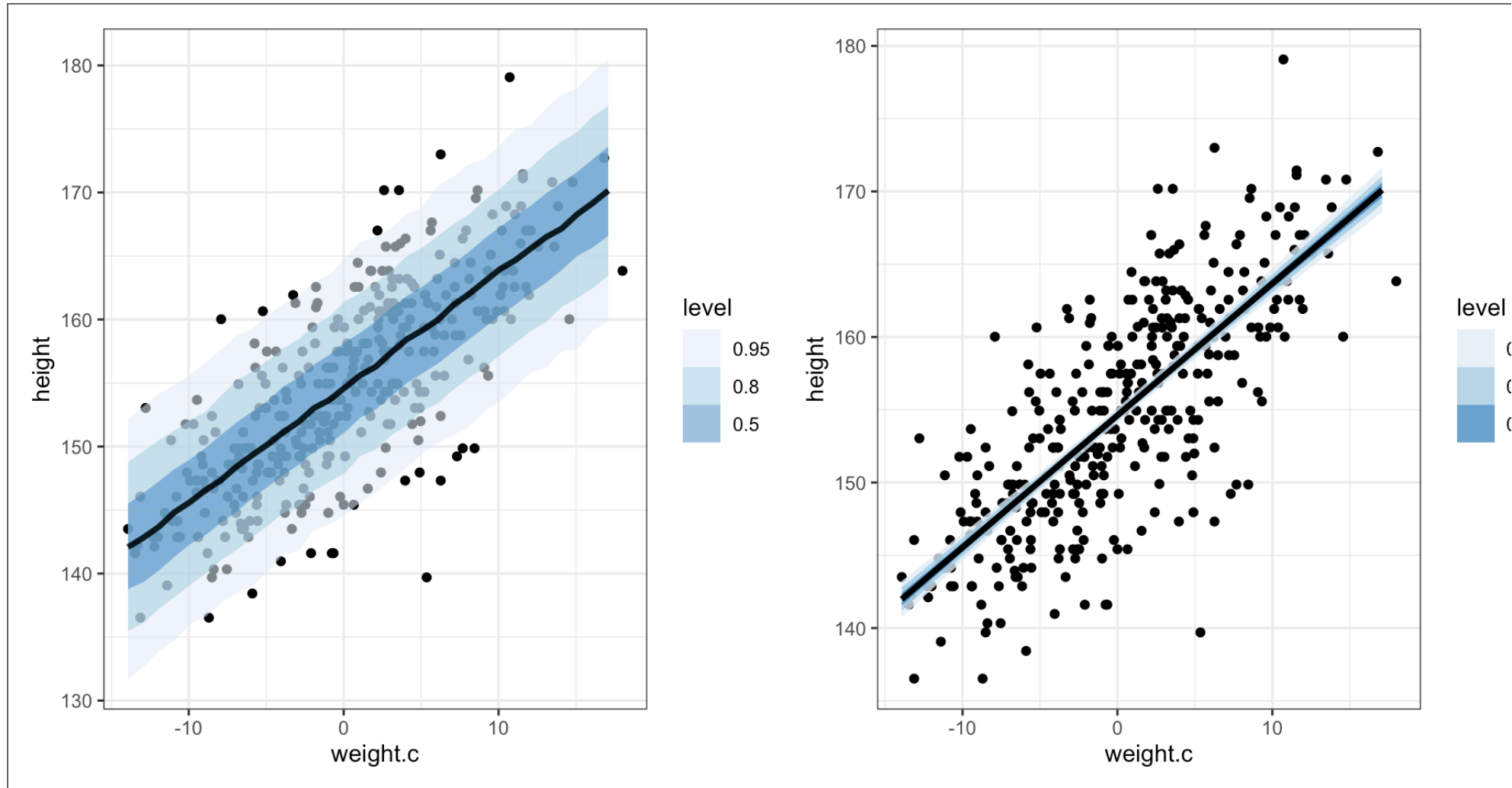
# Visualize both sigma uncertainty

```
1 ggplot(d2,  
2       aes(x = weight.c, y = height)) +  
3   geom_point() +  
4   # stat_lineribbon(data = mu, aes(x = weight.c, y = .epred)) +  
5   stat_lineribbon(data = sim_postpred, aes(x = weight.c, y = .prediction), alpha = 0.5) +  
6   scale_fill_brewer()
```





# Visualize both mu and sigma uncertainty



Speaker notes